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FEDERAL SPECIFICATION

LEATHER DRESSING, VESICANT GAS RESISTANT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers leather dressing suitable for field application to leather shoes to provide vesicant-gas-resistance.

1.2 Classification. The leather dressing shall be of one type.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein:

Federal Specifications:

C-N-200	-	Neat's-Foot Oil.
P-D-680	-	Dry Cleaning Solvent
VV-W-95	-	Wax, Paraffin, Technical.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Box, Fiberboard.
PPP-B-665	-	Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676	-	Boxes, Setup.
PPP-P-704	-	Pails, Metal: (Shipping, Steel, 1 Through 12 Gallon) and Soap-Products.

Federal Standards:

Fed. Test Method Std. No. 536 - Soap and Soap Products (Including Synthetic Detergents); Sampling and Testing.

Fed. Test Method No. 791 - Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 8030

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Military Specifications:

JAN-G-276 - Gum, Ester.
MIL-L-3122 - Leather, Cattlehide, for Footwear Uppers, Chrome-Tanned, Fat-Liquored.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials (ASTM) Standards:

D 87 - Melting Point of Petroleum Wax (Cooling Curve).
D 88 - Saybolt Viscosity.
D 93 - Flash Point by Pensky-Martens Closed Tester.
D 94 - Saponification Number by Color-Indicator Titration.
D 97 - Pour Point.
D 127 - Drop Melting Point of Petroleum Wax, Including Petrolatum.
D 217 - Cone Penetration of Lubricating Grease.
D 473 - Sediment in Crude and Fuel Oils by Extraction.
D 937 - Cone Penetration Petrolatum.
D 1500 - ASTM Color of Petroleum Products (ASTM Color Scale).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa., 19103).

3. REQUIREMENTS

3.1 Qualification. Leather Dressing furnished under this specification shall be a product that has been tested and has been listed on or approved for listing on the applicable qualified products list (see 6.5). Any change in the formulation of a qualified product will necessitate its requalification. The material supplied under the contract shall be identical, within manufacturing tolerances, to the product receiving qualification.

3.2 Composition. Leather dressing shall conform to the composition specified in table I.

TABLE I. Composition of leather dressing

	Percent by weight	
	Minimum	Maximum
Animal Oil	34.0	37.0
Ester gum	29.0	31.0
Paraffin wax	11.0	13.0
Microcrystalline wax	8.5	10.0
Solvent, drycleaning	11.0	15.0
Aluminum stearate	0.8	1.2

3.3 Ingredients. Ingredients used in the composition shall conform to the following requirements when tested as specified in 4.4 Blending of materials from different sources to meet the requirements for each ingredient will be permitted.

3.3.1 Animal oil. Animal oil shall conform to type I of C-N-200.

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3.3.2 Microcrystalline mineral wax. Mineral wax shall be microcrystalline petroleum wax, free from animal or vegetable waxes or other adulterants, and shall conform to table II.

TABLE II. Microcrystalline mineral wax

	Minimum	Maximum
Color A.S.T.M	--	5.0
Melting point (°C.)	74	85
Penetration (mm x 0.1 at 25°C.)	25	45
Viscosity at 210°F., saybolt		
Universal, seconds	60	80
Flash point (°C.)	240	--

3.3.3 Paraffin wax. Paraffin wax shall conform to type I, grade C of VV-W-95.

3.3.4 Ester gum. Ester gum shall conform to type II of JAN-G-276.

3.3.5 Aluminum stearate. Aluminum stearate shall conform to Table III.

TABLE III. Aluminum sterate

	Minimum	Maximum
	Percent	Percent
Aluminum (Al_2O_3)	7.5	11.0
Moisture	---	1.5
Free fatty acid	5.0	20.0

3.3.6 Dry cleaning solvent. Solvent shall conform to type II of P-D-680.

3.4 Finished leather dressing.

3.4.1 Form. The finished leather dressing shall be a stable, smooth, homogeneous mixture of creamy consistency prepared from materials specified herein, free from lumps, crystalline or grainy appearance, and shall have no objectionable odor. The leather dressing shall be suitable for treatment of leather by hand rubbing or by the hot-dip method and shall make the treated leather impermeable to vesicants.

3.4.2 Chemical and physical requirements. The finished dressing shall conform to the chemical and physical requirements specified in tables IV and V, respectively, when tested as specified in 4.5.

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TABLE IV. Chemical requirements of finished leather dressing

	Minimum percent	Maximum percent
Highly ionized acids[1]	--	None
Free fatty acids (expressed as oleic)	--	4.0
Unsaponifiable matter	22.0	28.0
Rosin acids (percent of total acids)	24.0	36.0
Alumina (Al_2O_3)	0.06	0.14
Sediment	--	0.3
Moisture	Shall pass test specified in 4.5.9	
Total volatile matter	11.0	15.0
Homogeneity	Shall pass test specified in 4.5.9	

[1]Initial pH for finished leather dressing shall be not less than 3.7.

TABLE V. Physical requirements of finished leather dressing

	Minimum	Maximum
Color A.S.T.M	--	5
Melting point (°C.)	46	56
Penetration (unworked at 25°C.)	180 mm.x0.1	220 mm.x0.1
Penetration (worked at 25°C.)	250 mm.x0.1	--
Flash point (°C.)	55	--

3.5 Vesicant gas resistance. Leather treated with the finished leather dressing shall be resistant to liquid mustard gas for at least 165 minutes when tested as specified in 4.5.11.

3.6 Workmanship. The leather dressing shall be produced in accordance with the best commercial practice and shall be free from any defects which may affect the serviceability, working properties, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein and for the testing for these requirements except for the vesicant gas resistance test which will be run by the Edgewood Arsenal. (This deviation applies only to conductance of the test by the supplier, not the need to comply with the vesicant gas resistance requirement). Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lotting. A lot shall consist of the leather dressing produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions provided the process is continuous. In the event that the process is a batch operation, each batch shall constitute a lot (see 6.2).

4.3 Sampling for test.

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4.3.1 Four-ounce cans or 8-ounce cans. When the contract or purchase order specifies 4 or 8 ounce cans (see 6.1), a total of 24 cans shall be selected at random from each lot of finished leather dressing.

4.3.2 Five-gallon cans. When the contract or purchase order specifies 5-gallon cans (see 6.1), sampling shall be conducted as follows:

TABLE VI. Sampling for 5-gallon cans

Lot size (5-gallon cans)	Sample size (5-gallon cans)
Up to 500	6
501 to 35,000	8
35,001 and over	12

A total specimen of 96 ounces shall be taken from the sample 5-gallon cans selected in accordance with table VI. An approximate equal number of ounces shall be removed from each 5-gallon can selected. The specimen taken from each can shall be divided equally and placed in two clean, dry containers. The containers shall be labeled to identify the can and the lot represented by the specimen.

4.4 Inspection.

4.4.1 Four-ounce cans or 8 ounce cans. From the sample selected in accordance with 4.3.1, a total of 12 cans shall be forwarded to the laboratory designated by the contracting officer (see 6.1) and tested for vesicant gas resistance (4.5.11). The other 12 cans shall be used for the remaining tests in 4.5. The lot represented by the sample shall be rejected if the sample fails to pass any of the tests specified.

4.4.2 Five-gallon cans. From the sample selected in accordance with 4.3.2, a total of 48 ounces, consisting of a specimen from each 5-gallon can selected, shall be forwarded to the laboratory designated by the contracting officer (see 6.1) and tested for vesicant gas resistance (4.5.11). The other 48 ounces shall be used for the remainder of the tests in 4.5. The lot represented by the sample shall be rejected if the sample fails to pass any of the tests specified.

4.5 Tests.

4.5.1 The methods specified in American Society for Testing and Material (ASTM Standards) shall be used for the following determinations:

	Method
Color, ASTM	D1500
Pour point	D97
Viscosity	D88
Penetration on dressing (worked and unworked)1/	D217
Penetration on wax	D937
Flash point	D93
Melting point (dressing)	D127
Melting point (wax)	D87
Sediment	D473
Saponification number	D94

1/ Penetration (unworked) shall be made on the dressing as furnished in the sample containers. A composite sample may be taken for all other tests.

4.5.2 Acidity. Weigh approximately 10 grams of material and extract with three 50 ml. portions of neutral 95 percent alcohol at 50° to 60° C. Combine the three extracts. Make electrometric titration by means of a glass electrode, using approximately 0.1 N alcoholic KOH (made with 95 percent ethyl alcohol), adding the KOH solution in 2 ml. portions until the expected endpoint is approached. Then continue addition in 1 ml. portions until approximately pH 11.0 is reached. Record pH (or e.m.f.) after each addition. Make a blank determination on reagents and apply necessary corrections. Plot ml. of KOH against pH and determine point (or points) of maximum slope (or break).

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4.5.2.1 Free fatty acids. The point of maximum slope or break on the curve for free fatty acids will be in the alkaline range. The ml. of KOH corresponding to this point represents amount equivalent to free fatty acids. Calculation of results is as follows:

$$\frac{\text{(ml. KOH) (normality) (28.2)}}{\text{gm. of sample}} = \text{percent oleic acid}$$

Report as percent fatty acids.

4.5.3 Unsaponifiable matter. Weigh accurately 3 grams of material into a 250 ml. Erlenmeyer flask. Add 50 ml. alcoholic (85 percent) 1.25 N KOH, attach a reflux condenser to the flask, and reflux gently for about 3 hours or until saponification is complete. Test for excess alkalinity after saponification by adding a few drops of phenolphthalein to the flask. Rinse into a 500 ml. separatory funnel with hot water. (Note: There must be no grease on the stopcock). Allow to cool. Add 100 ml. petroleum ether and swirl vigorously, but do not shake. Allow to settle and draw off all except approximately 1 ml. of the aqueous layer. Repeat this extraction of the soap solution three times with 50 ml. portions of petroleum ether. Save the extracted soap solution for determination of rosin acid specified in 4.5.4. Combine the petroleum ether extracts and wash with successive small portions of hot water until the washings are no longer alkaline to phenolphthalein. Add more petroleum ether, if necessary. Swirl the funnel so as to bring the last drop of water down to the stopcock and draw off until the ether solution fills the bore of the stopcock. Wipe out the stem of the funnel with a cotton swab. Draw off the ether solution (portionwise if necessary) into a weighed glass or porcelain dish. Evaporate the ether, place dish in an oven at 100° C. for 30 minutes, cool in a desiccator, and weigh. The increase in weight of the dish is unsaponifiable matter. Calculate the percent unsaponifiable matter.

4.5.4 Rosin acids. The aqueous soap solution obtained in the determination of unsaponifiable matter shall be used in the determination of rosin acids. The alcohol shall be boiled from the solution and the soap solution concentrated to approximately 500 ml. Place the soap solution in a 1000 ml. separatory funnel, add dilute hydrochloric acid (1:1) with frequent agitation until the fatty and rosin acids separate (usually forms a white cloudy mass). Extract the mixed acids with 100 ml. of petroleum ether. The aqueous and ether layers shall be shaken together until the water phase is almost clear. The ether layer shall be washed with hot water until the washings are neutral to methyl red. Filter the ether layer through a No. 42 Whatman filter paper, or equivalent, into a tared flask, preferably one which can be connected by a ground glass joint to a reflux condenser. Place the flask and contents on a steam bath to remove most of the petroleum ether and then dry to constant weight (0.001 gm.) in an oven at 100° to 105° C. The dried material represents the mixed acids. The percent of rosin acids (R) in the mixture shall be determined by method 701 of Fed. Test Method Std. No. 536.

4.5.5 Aluminum oxide. Ignite 1 to 2 gm. of the sample in a weighed platinum or nickel crucible until the carbon is removed. The residue is mixed with 2.5 to 3.0 gm. of a mixture of equal parts Na_2CO_3 and K_2CO_3 and fused. Place the crucible in a suitable vessel and dissolve the melt by digestion on the steam bath with dilute HCl (solution must be complete; if not, fusion was incomplete and the determination must be started again on new sample). Evaporate to dryness on the steam bath and continue heating in the oven at 120° C. for 30 minutes. Digest with 25 ml. 1:1 HCl (on steam bath) until residue dissolves or has had ample opportunity to dissolve. Add 100 ml. of water, filter, and wash residue. Continue the filtrate and washings, precipitate aluminum with a slight excess of ammonium hydroxide, filter, wash with 2-percent NH_4Cl solution, and ignite to constant weight. The residue, which should be white, is Al_2O_3. Report as percent Al_2O_3.

4.5.6 Sediment. The method for determining sediment shall be as specified in 4.5.1, except that a 5 gm. sample with carbon tetrachloride for extraction shall be used. Calculate the increase in weight as percent sediment.

4.5.7 Moisture in aluminum stearate. Weigh accurately 1 gm. of material in a wide-mouth weighing bottle, and heat in an oven at 105° C. to constant weight. Report loss in weight as percent moisture.

4.5.8 Moisture in finished leather dressing. Place approximately 3 grams of leather dressing in a porcelain dish, and heat with a low flame to the first appearance of smoke. The leather dressing shall not contain moisture in sufficient amount to cause spattering or bumping when heated. In the event of controversy, a quantitative test specified in method 3001 of Fed. Test Method Std. No. 791, shall be used.

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4.5.9 Homogeneity. Place leather dressing in a large test tube so that tube will be one-third to one-half filled. Melt dressing by immersing the test tube in a water bath at a temperature not exceeding 10°C. above its previously determined melting point. Dressing solution shall be homogeneous and shall show no evidence of solid particles.

4.5.10 Total volatile matter. Weigh approximately 5.0 gms. of leather dressing in a tared weighing dish. Heat at 110° ± 2°C. for 4 hours, with cover of dish removed, in an oven that will permit the elimination of the flammable solvent vapors safely. Cool in a desiccator (with cover on weighing dish) and weigh. Calculate total volatile contents as follows:

$$\text{Percent volatile matter} = \frac{\text{Loss in weight} \times 100}{\text{Original weight of sample}}$$

4.5.11 Vesicant gas resistance.

4.5.11.1 Preparation of leather dressing. The amount of leather dressing required and the size of container used must be sufficient for total immersion of the test samples of leather. Place a sufficient amount of leather dressing in a suitable container and heat to 150° to 170°F. A suitable means should be employed to maintain this temperature throughout the immersion of the leather test samples. Care should be exercised during this operation as the temperature of the leather dressing is approaching the flash point. When the leather dressing starts to melt, stir the dressing and mash up all lumps. Continue the heating and stirring until the dressing has completely melted. When the amber liquid has become crystal clear, it is hot enough to start dipping the leather test samples. The heat should be adjusted to keep the leather dressing clear at all times, however, it should not be overheated as this will exceed the fire point of the leather dressing.

4.5.11.2 Preparation of leather test samples. The leather used for the test shall be Leather, Cattlehide, For Footwear Uppers, Chrome-Tanned, Fat-Liquored, MIL-L-3122, type II, class 1, condition 1, treatment C. A piece of the leather approximately 2-1/2-inches by 9-inches shall be weighed and then dipped into the melted leather dressing (4.5.11.1) for approximately 2 minutes. After immersion, the test samples are drained for approximately 30 seconds. Remove the excess leather dressing by wiping with clean cotton rags or by rinsing in gasoline or fuel oil and then wiping with rags. Weigh the test sample and calculate the percent weight gain. Continue the above dipping process until the weight gain of the test sample is a minimum of 60 percent of the original weight.

4.5.11.3 Determination of resistance. Determine the mustard vapor resistance of the test samples using Edgewood Arsenal Q170 Static Diffusion Tester.

4.6 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table VII. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and AQL shall be 4.0 defects per hundred units.

TABLE VII. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose stapling. Distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.1).

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5.1.1 Level A.

5.1.1.1 Unit packaging. Four ounces of dressing shall be packaged in a round commercial can with friction slip cover provided with curled edges and special unlocking features. The unlocking feature may be a pressure pop-off lid, a mechanical locking device, or cutouts capable of receiving a coin or similar object for prying off the lid. Five gallons of dressing shall be furnished in a pail conforming to PPP-P-704, type II, class 1.

5.1.1.2 Intermediate packaging. Unless otherwise specified (see 6.1), 24, 4-ounce cans of dressing shall be intermediate packaged in a close-fitting box conforming to PPP-B-566, PPP-B-665, or PPP-B-676.

5.1.2 Level B.

5.1.2.1 Unit packaging. Four ounce and eight ounce quantities of dressing shall be furnished in a round metal commercial can with friction slip cover provided with curled edges and special unlocking features. The unlocking feature may be a pressure pop-off lid, a locking device, or cutouts capable of receiving a coin or similar object for prying off the lid. Five gallons of dressing shall be furnished in a pail conforming to PPP-P-704, type II, class 1.

5.1.2.2 Intermediate packaging. Twenty-four, 4-ounce or twelve 8-ounce cans of dressing shall be intermediate packaged in a close-fitting box conforming to PPP-B-566, PPP-B-665, or PPP-B-676.

5.1.3 Level C. The dressing shall be packaged to provide adequate protection against deterioration and damage from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Unless otherwise specified (see 6.1), four intermediate packages of 4-ounce cans of dressing shall be packed in a close-fitting box conforming to PPP-B-636, class weather-resistant, grade W5c. The box shall be closed in accordance with the appendix to the box specification. Five gallon of dressing require no additional packing.

5.2.2 Level B. Four intermediate packages of 4-ounce cans of dressing or six intermediate packages of 8-ounce cans of dressing shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. Five gallon pails of dressing require no additional packing.

5.2.3 Level C. Dressing, packaged as specified in 5.1.3 shall be packed to insure carrier acceptance and safe delivery at destination, in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to the markings required by the contract or order, the cans and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to markings required by the contract or order, the cans and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2.1 Special marking. All data shall be lithographed or printed and shall be clear, legible, non-fading, and durable.

5.3.2.2 Four-ounce and 8-ounce cans. Label data shall be placed on the cover and side of the can as specified below:

(a) Can cover. Label data shall be as follows:

4 oz. NET or 8 oz. NET as appropiate
FLAMMABLE
Leather Dressing, Vesicant Gas Resistant, M2.

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General Instructions: Remove lacings from shoes. Remove all loose dirt and mud. Best results obtained when dressing applied at warm temperature. Cold delays penetration. Use dressing generously. Rub in well. Specified quantities must be used. For further instructions, see side of can.

CAUTION: Do not use can or contents for food purposes.

(b) Side of can. Label data shall be as follows:

Detailed Instructions:

1. Stir contents thoroughly. Apply to outside and inside and tongue of shoes. Work into leather by kneading and flexing. Concentrate on all single thickness parts and seams. Allow shoes to stand for 3 hours in well-ventilated place. Combat boots (10½-inch high) require two 4-ounce cans and 20 to 30 minutes application. Other foot-wear in proportion to height.

2. Repeat above application (using one 4-ounce can) and replace lacings. Shoes can now be worn.

3. After at least 8 hours wear, reclean shoes and apply dressing (using one 4-ounce can) to outside of shoes only, working as much as possible into the leather.

4. Repeat direction 3 each day for next four days.

5. Apply dressing (one 4-ounce can) at least once a week thereafter.

5.3.2.3 Five-gallon pail. Label data shall be placed on the cover and side of the can as specified below:

(a) Can cover. Label data shall be as follows:

5 GALLON NET
FLAMMABLE
Leather Dressing, Vesicant Gas Resistant, M2.

(b) Side of can. Label data shall be as follows:

1. EQUIPMENT AND MATERIAL REQUIRED.

a. 30-gallon container (or larger).
b. Kitchen immersion heater (TM10-4540-201-15).
c. Fuel for operating the heater and for rinsing the boots.

2. PREPARATION OF LEATHER DRESSING.

a. Transfer the leather dressing from the 5-gallon pails into the 30-gallon container. Figure one pint of dressing for each pair of boots to be dipped plus ten additional gallons of dressing to fill the 30-gallon container to the minimum depth required for dipping.

b. Place the immersion heater in the 30-gallon container and light the heater.

c. Stir the dressing with a long stick when it starts to melt. Mash all lumps.

d. Continue heating and stirring until the amber liquid has become crystal clear.

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e. CAUTION: DO NOT OVERHEAT.

f. If the backs of the fingers burn when momentarily touching the outside of the container below the liquid level, or if the dressing smokes, shut off the heater and allow the container to cool somewhat.

g. To speed up the operation, a second 30-gallon container and immersion heater can be used for premelting the dressing.

(1) Place the unopened cans of dressing in the container and fill with water.

(2) Operate the heater to keep the water just below the boiling point.

3. METHOD OF APPLICATION.

a. Boots must be clean and free of all mud, debris, and other foreign matter.

b. Examine the boots for cracks or broken seams before dipping. Damaged boots are not satisfactory for dipping and should be replaced.

c. Remove boot laces and tie a lace to the top eyelet of each boot.

d. Immerse the boots one pair at a time completely in the molten dressing for two minutes, insuring that the dressing has filled the toes of the boots. Allow the laces to hang over the outside of the container.

e. After two minutes, remove the boots from the dressing (a stick may be used to handle the boots) and allow to drain over the container for half a minute.

f. While the boots are still hot, wipe with rags, preferably cotton. Pay particular attention to the inside especially the toe area, instep and tongue. This should require two hours of concentrated effort.

g. Boots can be cleaned more easily and effectively by five minutes of rinsing in fuel (ten immersions of one-half minute each where boots are emptied prior to each immersion).

h. Discard the fuel after 100 pairs have been rinsed and replenish with fresh fuel. When the air temperature is below 70°F, it is necessary to change fuel after 50 pairs have been rinsed.

i. Boots should be air dried for 24 hours prior to use. However, in an emergency, boots can be worn after cleaning.

j. After the boots have once been treated, no further application of dressing is necessary.

4. SAFETY PRECAUTIONS.

a. Care should be exercised when relighting the heaters to avoid sparking as the protective dressing is flammable when hot.

b. Have a cover available for the 30-gallon container to snuff out the flame if ignition should occur.

c. Hands and arms should be covered during the dipping operation to avoid splashing hot dressing on the bare skin.

d. Do not use can or contents for food purposes.

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6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Quantity of cans as required for sampling and inspection (see 4.3.1, 4.3.2, 4.4.1, and 4.4.2).

(c) Size of container (see 5.1).

(d) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

(e) When intermediate packages are not required for level A (see 5.1.1.2).

(f) When other than four intermediate packages are required for level A packing (see 5.2.1).

6.2 Batch. A batch is defined as the quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

6.3 The following procedure, although not mandatory, is recommended for the manufacture of the material. Add the aluminum stearate to the animal oil and stir vigorously until all of it is suspended in the oil. Heat to 125° C. and when the aluminum stearate has dissolved add the ester gum. (The ester gum preferably should be added in the size of 2-inch lumps or smaller. Avoid the use of powdered ester gum unless it is freshly pulverized. Powdered ester gum oxidizes rapidly and the oxidized material forms sticky insoluble residues and causes foaming). Continue the heating until the temperature of the mixture reaches 200° to 230° C. Heat the mixture at this temperature for 1 hour after the addition of the gum. Discontinue the heating. Continue the stirring and gradually add the waxes in small pieces at a rate such that the wax is melted without interfering with the agitator. Maintain the temperature of the mix above 125° C. during the addition of the waxes to insure rapid dissolving and uniform distribution in the mixture. After the addition of the waxes is complete, gradually add the solvent to the mixture. Agitate the mixture for 5 to 10 minutes maintaining the temperature at 100° to 105° C. to insure uniform distribution of the solvent. Run the liquid mixture at a temperature of 100° to 105° C. from the reaction kettle to a pony mixer or other device fitted with a similar stirring mechanism. Continue the stirring and cool to 40° to 45° C. At this point, transfer the material to the containers. Avoid heating the creamed material during the transfer to containers.

6.4 This specification incorporates the requirements of MIL-D-51A dated September 5, 1950 and Amendment-1 dated February 18, 1953.

6.5 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Standardization Division, General Services Administration, Washington, D.C., 20406, and information pertaining to qualification of products may be obtained from that Activity.

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